Effects of Urban Infrastructure on Violent Crime & Pedestrian-Vehicle Accidents

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Introduction

We present two machine learning models to predict violent crime in an area (regression) and classify an area as a pedestrian-vehicle accident (PVA) hotspot as a function of the quality and quantity of public infrastructure/institutions (e.g. number of parks, number of speed humps, etc) that a Charlotte census tract has. A census tract is a region of analysis used by the US Census bureau which estimates neighborhoods. Census tracts are a preferred unit of analysis over ZIP codes because census tracts are more granular and are designed to be demographically homogeneous. Our intended goal in this project is not simply to make predictive models, but instead to examine parameter values, patterns, and p-values to observe which features are associated with violent crime and PVA hotspots.

Previous research has been done to analyze the effects of places of worship (POWs) on crime rates. Environmental criminologists often cite that areas with a high amount of non-residential activity (shopping, attending school, attending a POW) are more likely to have higher crime rates. Stores, schools, places of worship, etc. have the ability to increase traffic to an area, causing targets to become more susceptible to criminal activity (Wo, 2023). Wo found that even when controlling for other "crime-generating" factors (poverty rate, number of alcohol vendors, etc.), that the number of POWs an area has is positively correlated with both property crime and violent crime (Wo, 2023). Wo's study was limited to only analyzing POWs with data from Washington, D.C (Wo, 2023). In this study, we aim to apply Wo's methods to other types of urban infrastructure and institutions in Charlotte, NC.

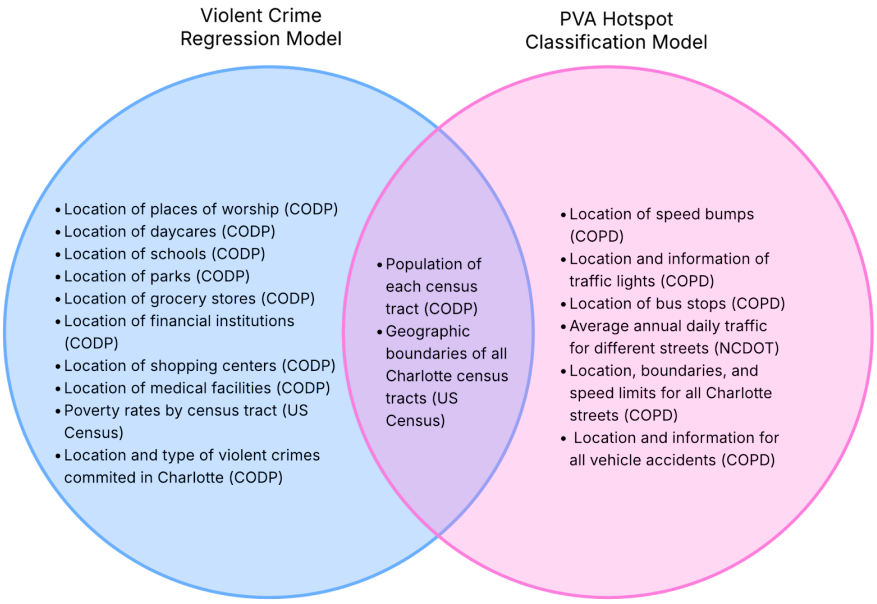
Additionally, a study by Tester et al. (2004) used multiple regression analysis with a logistic regression model to see if the presence of speed humps in Oakland, CA neighborhoods could be used to predict child PVAs. Tester et al. (2004) found that speed humps decreased the log odds of a child being struck within their neighborhood in Oakland, CA (Tester et al., 2004). We aim to extend Tester et al.'s research to a different geographic region (Charlotte, NC) and also examine other useful variables in classifying a census tract as a PVA hotspot.

Data Collection, Cleaning, & Preparation

The data used to train our models came from the City of Charlotte Open Data Portal (CODP), the US Census Bureau, and the North Carolina Department of Transportation (NCDOT). The CODP offers multiple geo-spatial datasets that can be downloaded as a GEOJSON file. We obtained shape files for census tracts from the US Census Bureau. Using these shape files, polygons (geographical regions bounded by coordinates) could be constructed for each Charlotte census tract. There are 239 census tracts within Charlotte city limits; this number was calculated by observing if the centroid of a census tract polygon lies within the Charlotte city limits polygon..

From the census tract polygons, a spatial-join can be performed on a set of coordinate points to locate which Charlotte census tract a point lies within. This allowed us to merge multiple datasets from the Charlotte Open Data Portal and NCDOT for aggregation grouped by census tract.

Charlotte/NC Datasets Utilized & Merged

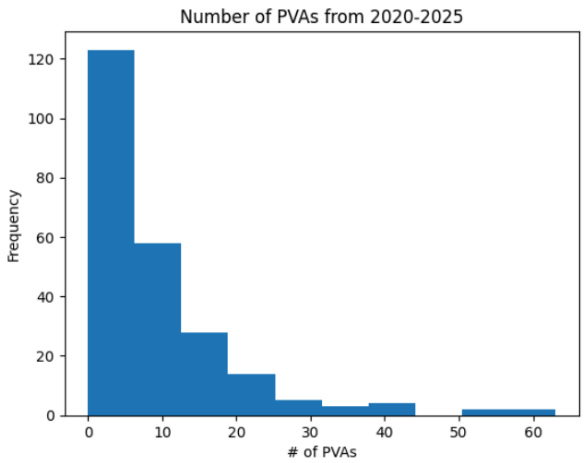


CODP signifies the dataset came from Charlotte Open Data Portal. NCDOT signifies the dataset came from the North Carolina Department of Transportation. US Census signifies the dataset originated from the US census bureau.

After merging each datapoint in each dataset with its respective census tract, census tract counts were aggregated across all datasets, yielding features like 'number of daycares' and 'number of speed humps'. The dataset of car crashes was filtered to include only pedestrian involved crashes. Both target datasets, car crashes and violent crimes, were filtered to only include data from 2020 onward to reduce look ahead bias.

We originally considered dividing all feature counts by the population/average daily traffic of their respective census tracts, however we decided to simply include population/average daily traffic as a feature to allow our models to understand the relationship between them.

Given that we had access to how many PVAs a census tract has had since 2020 and that we wanted this model to be a classification model. We discretized PVA counts for each census tract by marking the census tract as a positive if the PVA counts for that census tract were greater than or equal to the median value of all PVA counts, negative otherwise. We chose to use the median instead of the mean since the distribution of PVAs was heavily skewed right, as can be seen in the histogram below.



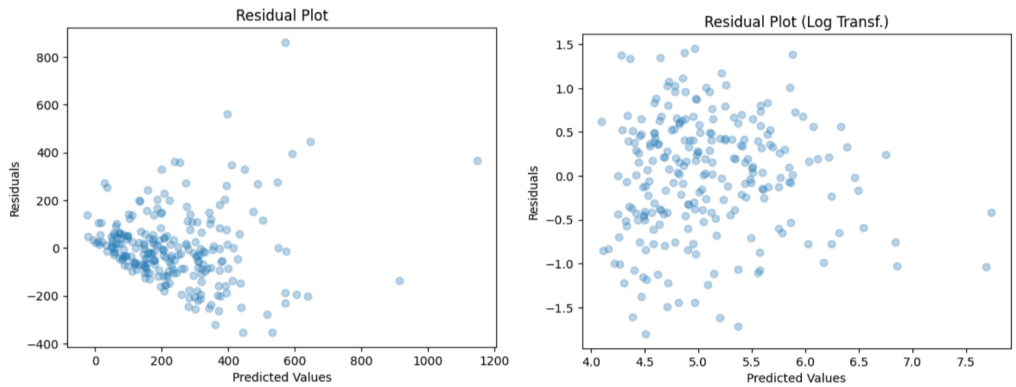
Histogram of PVA count frequencies showing right skewness

Regression Model

We decided to regress the number of violent crimes in a census tract as a function of: population, poverty rate, the total number of schools the tract has, the number of worship centers the tract has, the number of parks the tract has, the number of financial institutions the tract has, the number of medical facilities the tract has, the number of daycares the tract has, and the number of shopping centers the tract has. Since this is largely an exploratory study and constructing the most accurate model is not our number one priority, we did not perform feature selection, and instead chose to include variables of interest along with control variables like population and poverty rate.

Variance inflation factors were calculated for all of our features, yielding values of less than two, suggesting that there are no multicollinearity problems with our data.

We tested three different regression models, ordinary least squares (OLS), poisson regression, and negative binomial regression with population as an exposure term (as suggested by Wo for count data that is overdispersed, meaning the variance is greater than the mean). We compared the models based off of their residual plots as well as root mean squared error (RMSE). Prior to fitting each of the models, we standardized all of the features (using z-scores) to be on the same scale. OLS yielded an RMSE of 162, Poisson regression yielded 180, and negative binomial regression yielded 221. All the models shared a very similar relationship between predicted values and residuals– residuals were evenly spread around 0 and the variance of residuals became higher as the predicted value became higher (heteroskedasticity). Given that OLS had the highest RMSE, is the easiest to interpret, and the residual plots all looked similar, we chose OLS as our regression model. We then decided to apply a log transformation (base e) to the target variable (number of violent crimes) to better match the assumptions of linear regression and also to allow for better coefficient interpretability as coefficients can be interpreted as the expected % change in the dependent variable for a one unit increase in the independent variable.



Left: OLS model without log transformation on target variable
Right: OLS model with log transformation on target variable

After training the OLS with the log of violent crimes as the response, we obtained the following output:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         violent_crimes   R-squared:                       0.464
Model:                            OLS   Adj. R-squared:                  0.443
Method:                 Least Squares   F-statistic:                     22.04
Date:                Tue, 29 Apr 2025   Prob (F-statistic):           9.95e-27
Time:                        00:24:30   Log-Likelihood:                -241.91
No. Observations:                 239   AIC:                             503.8
Df Residuals:                     229   BIC:                             538.6
Df Model:                           9
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              5.0851      0.044    115.576      0.000       4.998       5.172
Population         0.0862      0.048      1.803      0.073      -0.008       0.180
pct_poverty        0.3906      0.047      8.337      0.000       0.298       0.483
TotalSchools       0.0236      0.050      0.476      0.635      -0.074       0.122
worship_centers    0.1133      0.055      2.054      0.041       0.005       0.222
parks              0.0831      0.052      1.594      0.112      -0.020       0.186
financial_insts    0.0623      0.055      1.135      0.258      -0.046       0.170
med_facilities    -0.0127      0.049     -0.258      0.797      -0.110       0.084
daycares           0.2001      0.057      3.526      0.001       0.088       0.312
shopping_centers   0.1139      0.052      2.182      0.030       0.011       0.217
```

The low F-statistic tells us that our model is statistically significantly useful for modeling violent crimes versus a baseline model that always predicts the mean number of violent crimes. The coefficient of population suggests that holding all other variables constant, for every one standard deviation increase in population (recall we scaled features using z-scores), we can expect an ~8% increase in the expected number of violent crimes, however this is not statistically significant (at alpha=0.05). This is consistent with published research that the population of an area is not a good estimate of the 'at-risk' population, but rather the ambient population ("foot-traffic") is (Andresen, 2010). As expected, the poverty rate is positively correlated with violent crime at a statistically significant level. The number of schools in an area seems to have a negligible effect (p-value > 0.05) on violent crime. We did find similar results as Wo in his study, finding that holding all other variables constant (including strong predictors like poverty rate), a one standard deviation increase in POWs leads to an expected ~11% increase in violent crime (Wo, 2023). The low p-value for the relationship between POWs and violent crime confirms Wo's previous findings as well. The number of parks, financial institutions, and medical facilities a

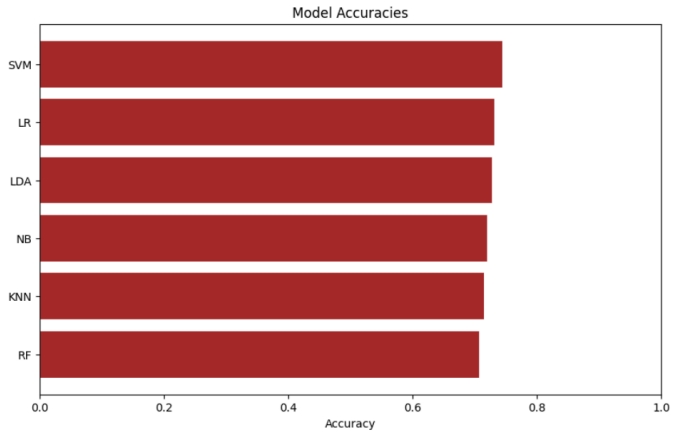
census tract has did not have statistically significant effects on violent crime given the presence of the other predictors. We found the number of daycares to be one of the strongest predictors of violent crime that we tested. A one standard deviation increase in the number of daycares a tract has (controlling for all other variables) corresponds to an expected ~20% increase in violent crime. Shopping centers are also statistically significantly positively correlated with violent crime. Controlling for all other variables, a one standard deviation increase in the number of shopping centers is expected to result in a ~11% increase in violent crime.

Classification Model

We attempted to classify whether or not a census tract was a PVA hotspot (binary classification). We defined a PVA hotspot to be a census tract with a number of PVAs (between the years 2020-2025) that is higher than the median across all census tracts. The features we started with to classify PVA hotspots were: the number of bus stops that a census tract has, the number of traffic lights a census tract has, the number of speed humps a census tract has, the percentage of traffic lights that have pedestrian crosswalk buttons in a census tract, the population density of a census tract, the estimated average daily traffic that a tract receives, and the weighted speed limit average which is calculated as:

$$Weighted\ Speed\ Limit_j = \frac{\sum_{i=1}^{n} SpeedLimit_{i,j} * LengthOfStreet_{i,j}}{\sum_{i=1}^{n} LengthOfStreet_{i,j}}\ for\ ith\ street\ in\ jth\ census\ tract$$

We decided to do model selection and feature selection simultaneously. We tested six different machine learning algorithms: support vector machines (SVM), logistic regression (LR), linear discriminant analysis (LDA), naive bayes (NB), k-nearest neighbors (KNN), and random forest (RF). For each model we tested all possible combinations of our features and selected the combination that yielded the highest eight fold cross-validated accuracy (features were scaled for SVM, LR, and KNN). Then, a grid search was performed over different model hyperparameters and the combination of hyperparameters yielding the highest eight fold cross validated accuracy was selected. Lastly, all the models were compared on their fully optimized accuracy yielding the following results:



The upper limit for model accuracy was ~75%. Given that logistic regression was the second most accurate and is one of the most interpretable classification algorithms, we chose logistic regression as our final model. After fitting the logistic regression model, we obtained the following results:

```
                           Logit Regression Results
==============================================================================
Dep. Variable:        high_crash_rate   No. Observations:                  239
Model:                          Logit   Df Residuals:                      233
Method:                           MLE   Df Model:                            5
Date:                Tue, 29 Apr 2025   Pseudo R-squ.:                  0.2429
Time:                        15:41:33   Log-Likelihood:                -123.89
converged:                       True   LL-Null:                       -163.65
Covariance Type:            nonrobust   LLR p-value:                 1.063e-15
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                 0.4889      0.167      2.925      0.003       0.161       0.816
bus_stops             1.4483      0.254      5.704      0.000       0.951       1.946
daily_traffic         0.2185      0.155      1.413      0.158      -0.084       0.521
speed_bumps          -0.2151      0.156     -1.383      0.167      -0.520       0.090
pct_tl_crosswalk      0.4896      0.157      3.113      0.002       0.181       0.798
population_density    0.5713      0.221      2.582      0.010       0.138       1.005
=====================================================================================
```

Logistic regression output. Pct_tl_crosswalk refers to the percentage of traffic lights that have pedestrians crossing buttons.

We found the number of bus stops in an area to be the best predictor of PVA hotspots. Controlling for other variables, a one standard deviation increase (features are scaled by z-score) in bus stops corresponds to a 1.4483 increase in the log odds that a census tract is a PVA hotspot. This correlation is also significant (p-value = 0.00). While this may suggest that bus stops increase pedestrian activity which can in turn increase pedestrian deaths, we do not have access to bus ridership data to control for in our model. Average daily traffic is positively associated with an area being a PVA hotspot but not at the statistically significant level. Keeping all other variables constant, speed humps do have a negative effect on the log odds that an area is a PVA hotspot, but this correlation is not statistically significant and warrants further research. Controlling for other variables, the percentage of traffic lights with pedestrian crossing buttons is statistically significantly positively correlated with the log odds that an area is a PVA hotspot. Every one standard deviation increase in the percentage of this variable corresponds with a 0.4896 increase in the log odds that an area is a PVA hotspot.

Our findings for the effects of bus stops and crosswalks echo findings from Anis, Geedipally, and Lord (2025). Geedipally and Lord (2025) found that one of the biggest risk factors for PVAs is the combination of bus stop quantity and their proximity to intersections, as these present the biggest risk for pedestrians trying to board/depart from buses.

Conclusion & Limitations

This study applied different machine learning algorithms to understand and explain the complex relationships between the quality/quantity of public infrastructure and the prevalence of violent crime and PVAs within a geographic region. In our regression analysis of violent crimes, we were able to confirm Wo's previous findings that POWs do seem to be positively correlated with violent crime, even when controlling for known predictors of crime such as population and poverty rate. Additionally, we found the number of shopping centers and the number of daycares in a census tract to be positively correlated with violent crime (given the presence of the other controlling variables). In regards to PVA hotspots, we found that the number of bus stops, the percentage

of traffic lights with pedestrian crossing buttons, and the population density of a census tract are all positively correlated with the log odds that an area is a PVA hotspot. We did not achieve the level of significance needed to confirm the ability of speed humps to reduce PVAs, as reported by Tester et al. (2004). However, there are limitations within our study and methods. (1) This study was a cross-sectional study of multiple geographic regions at one point in time. To more rigorously test associations, one should conduct longitudinal studies that follow geographic regions over time. (2) There may be confounding variables we did not control for in our regression analysis. For example, if we had access to bus ridership data, we would be able to more accurately test if the proximity of bus stops alone is a good indicator of a PVA hotspot. (3) Our study was limited to just 239 census tracts located only within Charlotte, NC. This means that results from our study should not be applied to other geographic regions without further testing. (4) Our study of violent crimes grouped all types of violent crime together. The dynamics of violent crime are complex and the factors that affect assault may be different from the factors that affect homicides.

References

Andresen, M., Jenion, G. Ambient populations and the calculation of crime rates and risk. Secur J 23, 114–133 (2010). https://doi.org/10.1057/sj.2008.1

Anis, M., Geedipally, S. R., & Lord, D. (2025). Pedestrian crash causation analysis near bus stops: Insights from random parameters NB-Lindley models [Preprint]. arXiv. https://arxiv.org/abs/2410.22253

"City of Charlotte Open Data Portal." City of Charlotte Open Data Portal, City of Charlotte Government, data.charlottenc.gov/. Accessed 1 May 2025.

Tester, J. M., Rutherford, G. W., Wald, Z., & Rutherford, M. W. (2004). A matched case-control study evaluating the effectiveness of speed humps in reducing child pedestrian injuries. *American journal of public health*, *94*(4), 646–650. https://doi.org/10.2105/ajph.94.4.646

U.S. Census Bureau, U.S. Department of Commerce. (2023). Poverty Status in the Past 12 Months by Age. American Community Survey, ACS 5-Year Estimates Detailed Tables, Table B17020. Retrieved May 1, 2025, from https://data.census.gov/table/ACSDT5Y2023.B17020?q=poverty+rate&g=050XX00US37119$1400000.

U.S. Census Bureau. (2024). 2024 TIGER/Line® Shapefiles. Retrieved May 1, 2025, from https://www.census.gov/cgi-bin/geo/shapefiles/index.php?year=2024&layergroup=Census+Tracts

Wo J. C. (2023). Crime generators or social capital organizations? Examining the effects of places of worship on neighborhood crime. PloS one, 18(3), e0282196. https://doi.org/10.1371/journal.pone.0282196

Appendix

Function that takes in a model and finds the best features and hyperparameters based on cross-validated accuracy.

```
def subset_and_cv(model, features, target, cv, scale_data, param_grid) -> tuple:
  """
  Evaluates all possible subsets of features and selects the best subset based on
  cross-validated accuracy.
  Returns:
  tuple:
      - highest_f1 (float): The highest cross-validated F1-score achieved.
      - best_subset (tuple): The subset of feAature names that yielded the highest F1-score.
      - scaler (StandardScaler or None): The fitted StandardScaler object if `scale_data` is True, otherwise None.
  """
  if scale_data:
    scaler = StandardScaler()
    scaled_features = pd.DataFrame(scaler.fit_transform(features), columns=features.columns)
  else:
    scaler = None
    scaled_features = features
  feature_names = features.columns
  highest_acc = 0 # highest accuracy found
  best_subset = None
  # try all # of features
  for i in range(3, 6):
    for combination in combinations(feature_names, i):
      combination_df = scaled_features[list(combination)]
      cv_results = cross_validate(
          model, combination_df, target, cv=cv,
      )
      acc = np.mean(cv_results["test_score"])
      if acc > highest_acc:
        highest_acc = acc
        best_subset = list(combination) # update best subset seen so far
  # once we find best subset of features, we can optimize the model with gridsearch
  grid_search_obj = GridSearchCV(model, param_grid, cv=cv, scoring="accuracy")
  grid_search_obj.fit(scaled_features[best_subset], target)
  highest_acc = grid_search_obj.best_score_
  return highest_acc, best_subset, scaler, grid_search_obj.best_estimator_, grid_search_obj.best_params_
```

Example showing how infrastructure aggregate counts were calculated for each census tract.

```
bus_stops_df = bus_stops_df.to_crs(tracts_df.crs)
bus_stops_df = gpd.sjoin(bus_stops_df, tracts_df, how="left", predicate="within")
bus_stops_df = bus_stops_df.groupby('TractID').size().reset_index(name='bus_stops')
```

Code showcasing how merging was done for the regression model.

```
dfs = [
    schools_df, worship_df, libraries_df
    parks_df, grocery_store_df, financial_df, medical_fac_df,
    daycare_df, post_office_df, shopping_centers_df, incidents_df
]
full_crime_df = tracts_df.copy()
for df in dfs:
  full_crime_df = pd.merge(full_crime_df, df, on="TractID", how="outer")
full_crime_df = full_crime_df.fillna(0).set_index("TractID") # replace non-matched zip code values with 0
```

Code detailing how violent crime linear regression was fit.

```
scaler = StandardScaler()
scaled_features = pd.DataFrame(scaler.fit_transform(full_crime_df_train),
                               columns=full_crime_df_train.columns,
                               index=full_crime_df_train.index)
scaled_features = sm.add_constant(scaled_features)
model = sm.OLS(np.log(full_crime_df["violent_crimes"]), scaled_features)
model = model.fit()
print(model.summary())
```

Code detailing how discretization of PVAs was done.

```
# discretize the number of crashes to above median or below median
full_crash_df["high_crash_rate"] = full_crash_df.apply(
    lambda row: 1 if row["num_crashes"] >= full_crash_df["num_crashes"].median() else 0,
    axis=1
)
```